Setting up your home network for work

Article KB0010095 | Owner: Network Support | Subcategory: General | Applies to: Windows 11 laptops (Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840), macOS 14/15 MacBook Pro/Air, Mobile devices (iOS/Android) using Microsoft Authenticator, Home routers and modems

Summary

This article guides Acme Global employees on preparing their home network for reliable remote work. It covers checking ISP service, configuring routers for VPN and Wi-Fi, securing the network, and verifying connectivity to corporate resources. Follow the self-help steps to resolve common issues before contacting the Service Desk.

Symptoms

- Unable to connect to the GlobalProtect VPN client - error "Connection timed out"
- Frequent VPN disconnections every 5-15 minutes
- High latency or packet loss when accessing internal sites (ping >100 ms)
- Wi-Fi shows limited or no internet access despite router lights on
- Speed test results far below subscribed plan (e.g., <10 Mbps download)

Common causes

- ISP service interruption or throttling
- Outdated router firmware
- Incorrect WAN/LAN settings (double NAT, wrong VLAN)
- QoS or bandwidth-limiting features interfering with VPN traffic
- Wi-Fi channel congestion or weak signal
- Misconfigured DNS causing internal name resolution failures

Pre-checks

Before making changes, verify that the problem is not isolated to a single device. Connect a laptop or smartphone directly to the modem (if possible) and test internet access. Note the results of a speed test (e.g., speedtest.acme.example) and a ping to 8.8.8.8. If the modem alone works, the issue lies with the home router or Wi-Fi setup.

Self-Help Procedure: Optimising Your Home Network for Work

Follow these steps to prepare your home network for reliable VPN and application performance. Each step assumes you have administrative access to your router's web interface (usually http://192.168.1.1 or http://192.168.0.1).

1. 1. Update router firmware - log in to the router, locate the Firmware Upgrade section, and install the latest version from the manufacturer's website.
2. 2. Set the WAN connection to obtain IP address automatically (DHCP) unless your ISP requires PPPoE; note any VLAN ID supplied by the ISP and enter it in the WAN settings.
3. 3. Disable any double NAT: if you have a modem-router combo from the ISP set to bridge mode, or place your personal router in AP mode and let the ISP device handle routing.
4. 4. Enable VPN passthrough (often called IPsec Passthrough or PPTP Passthrough) and ensure UDP ports 4500 and 500 are open for GlobalProtect.
5. 5. Configure QoS: prioritize traffic to the IP address of your work laptop or mark the GlobalProtect application as high priority.
6. 6. Choose a less congested Wi-Fi channel (use a Wi-Fi analyzer app) and set the SSID to "ACME-Home" with WPA2-PSK AES encryption.
7. 7. Set a static DHCP reservation for your work laptop so it always receives the same internal IP, simplifying port-forwarding if needed.
8. 8. Save settings, reboot the router, and reconnect your devices.

Verification

After completing the self-help steps, test the connection to corporate resources. Launch the GlobalProtect client and

verify it connects without errors. Open a browser and navigate to https://it.acme.example to confirm the self-service portal loads. Run a ping to an internal server (e.g., file.acme.example) and check for <30 ms latency and 0% packet loss. Finally, perform a speed test and ensure download/upload speeds meet at least 50% of your contracted plan.

Service-Desk Agent Procedure

When a user reports home-network-related connectivity issues, follow this workflow to gather information, perform remote diagnostics, and determine whether escalation to Network Support is required.

1. 1. Confirm the user's identity and device details (laptop model, OS version, MAC address).
2. 2. Ask the user to run a speed test via https://speedtest.acme.example and record download/upload/upload and ping.
3. 3. Instruct the user to connect the laptop directly to the modem (bypassing the router) and retest the VPN connection; note if the issue persists.
4. 4. If the VPN works directly but fails through the router, request the router make/model, firmware version, and a screenshot of the WAN and LAN settings pages.
5. 5. Run a remote traceroute (using the user's laptop) to vpn.acme.example and capture any hops with >100 ms latency or packet loss.
6. 6. Review the GlobalProtect logs on the laptop for error codes (e.g., -120, -122) and share them with the Network Support team if needed.
7. 7. If all checks point to the ISP or home equipment, advise the user to contact their ISP or replace faulty hardware; otherwise, escalate to Network Support.

When to escalate

Escalate to the Network Support team when: (a) the user confirms the VPN works when connected directly to the modem but fails through their home router, and (b) the router firmware is up-to-date, VPN passthrough is enabled, and no double NAT is present. Include in the ticket: user name, employee ID, device hostname, router make/model/firmware, WAN connection type, speed-test results, traceroute output, and any relevant GlobalProtect error logs. Mark the ticket as P2 High if the user cannot access any corporate resources; otherwise P3 Moderate.

Keywords: home network setup, VPN troubleshooting, GlobalProtect, router configuration, Wi-Fi optimization, QoS, double NAT, firmware update, speed test, traceroute, Network Support